

ASSIGNMENT – 9

ADVANCED PROGRAMMING

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```
import java.util.*;

class Account {
    private int accountNumber;
    private String ownerName;
    private double balance;

    public Account() {
        this(accountNumber: 0, ownerName: "Unknown", balance: 0.0);
    }

    public Account(int accountNumber, String ownerName, double balance) {
        this.accountNumber = accountNumber;
        this.ownerName = ownerName;
        setBalance(balance);
    }

    public int getAccountNumber() {
        return accountNumber;
    }

    public void setAccountNumber(int accountNumber) {
        this.accountNumber = accountNumber;
    }

    public String getOwnerName() {
        return ownerName;
    }

    public void setOwnerName(String ownerName) {
        this.ownerName = ownerName;
    }

    public double getBalance() {
        return balance;
    }
}
```

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38     public void setBalance(double balance) {
39         if (balance < 0) {
40             throw new IllegalArgumentException(s: "Balance cannot be negative!");
41         }
42         this.balance = balance;
43     }
44
45     public void deposit(double amount) {
46         if (amount <= 0) {
47             throw new IllegalArgumentException(s: "Deposit amount must be positive!");
48         }
49         balance += amount;
50     }
51
52     public void withdraw(double amount) {
53         if (amount <= 0) {
54             throw new IllegalArgumentException(s: "Withdrawal must be positive!");
55         }
56         if (amount > balance) {
57             throw new IllegalArgumentException(s: "Insufficient balance!");
58         }
59         balance -= amount;
60     }
61
62     public void display() {
63         System.out.println("Account Number: " + accountNumber);
64         System.out.println("Owner Name: " + ownerName);
65         System.out.println("Balance: " + balance);
66     }
67 }
68
69 class SavingsAccount extends Account {
70     private double interestRate;
71
72     public SavingsAccount(int accountNumber, String ownerName, double balance, double interestRate) {

```

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class SavingsAccount extends Account {
    }

    public double calculateInterest() {
        return getBalance() * interestRate / 100;
    }

    @Override
    public void display() {
        super.display();
        System.out.println("Interest Rate: " + interestRate + "%");
        System.out.println("Interest Earned: " + calculateInterest());
    }
}

class CurrentAccount extends Account {
    private double overdraftLimit;

    public CurrentAccount(int accountNumber, String ownerName, double balance, double overdraftLimit) {
        super(accountNumber, ownerName, balance);
        this.overdraftLimit = overdraftLimit;
    }

    @Override
    public void withdraw(double amount) {
        if (amount <= 0) {
            throw new IllegalArgumentException(s: "Withdrawal must be positive!");
        }
        if (amount > getBalance() + overdraftLimit) {
            throw new IllegalArgumentException(s: "Overdraft limit exceeded!");
        }
        setBalance(getBalance() - amount);
    }

    @Override
    public void display() {
        super.display();
    }
}

```

```

    public void display() {
        super.display();
        System.out.println("Overdraft Limit: " + overdraftLimit);
    }
}

public class BankingsSystem {
    Run main | Debug main | Run | Debug
    public static void main(String[] args) {

        List<Account> accounts = new ArrayList<>();

        accounts.add(new SavingsAccount(accountNumber: 101, ownerName: "Alice", balance: 5000, interestRat... 5));
        accounts.add(new CurrentAccount(accountNumber: 102, ownerName: "Bob", balance: 3000, overdraftLi... 2000));

        for (Account acc : accounts) {
            System.out.println(x: "\n--- Account Details ---");
            acc.display();
        }

        try {
            accounts.get(index: 0).withdraw(amount: 10000);
        } catch (Exception e) {
            System.out.println("\nError: " + e.getMessage());
        }
    }
}

```

OUTPUT-

```

c:\Users\VITU\OneDrive\Desktop\java>cd "c:\Users\VITU\OneDrive\Desktop\java\" && javac BankingsSystem.java && java BankingsSystem
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--- Account Details ---
Account Number: 101
Owner Name: Alice
Balance: 5000.0
Interest Rate: 5.0%
Interest Earned: 250.0

--- Account Details ---
Account Number: 102
Owner Name: Bob
Balance: 3000.0
Overdraft Limit: 2000.0

Error: Insufficient balance!

c:\Users\VITU\OneDrive\Desktop\java>

```